Computer Science GCE A Level 2023 - Paper 2

Duration: 2h30 | Cameroon GCE Board

General Certificate of Education Board - General Certificate of Education Examination. 0795 Computer Science 2, ADVANCED LEVEL, June 2023. Duration: two and a half hours.

Answer any SIX questions. All questions carry 17 marks each. For your guidance, the approximate mark for each part of a question is indicated in brackets. You are reminded of the necessity for good English and orderly presentation in your answers. In calculations, you are advised to show all the steps in your working, giving your answer at each stage.

Exercise 1 (17 marks)

a. Which of direct access and sequential access takes a longer time? Explain your answer.

b(i). A double-sided DVD has a capacity of 8.5GB. How many CDs of 700MB storage capacity are required to store information in a full double-sided DVD?

b(ii)(a). We want to build a circuit that can be used to add two 4-bit numbers. We have full adders and half adders to use as components.

Explain the difference between a half-adder and a full-adder.

b(ii)(b). We want to build a circuit that can be used to add two 4-bit numbers. We have full adders and half adders to use as components.

Draw a block diagram of your circuit which shows how the numbers 1011 and 1110 are being added. Show and label all inputs and outputs. Use at least one half-adder and at least one full-adder.

b(iii). Describe TWO characteristics of a CPU that can affect the overall performance of a computer.

Exercise 2 (17 marks)

i(a). Consider the following 8-bit binary integer which is in two's complement form.

0	**1**	**1**	**1**	**1**	**0**	**1**	**0**

Give the denary equivalent of this binary integer.

i(b). Consider the following 8-bit binary integer which is in two's complement form: 0 1 1 1 1 0 1 0.

Show how this binary integer would be represented in hexadecimal.

i(c). Consider the following 8-bit binary integer which is in two's complement form: 0 1 1 1 1 0 1 0.

Show the result obtained when this binary integer is added to itself. State its decimal equivalent and explain the result.

ii(a). A RISC processor has a 32-bit architecture, with instructions that are 1-word long. The processor has 64 registers each of which is 32 bits long and supports 45 instructions (operations). Besides the operator field which holds the operation code (Opcode), each instruction has an immediate operand and two register operands. The Opcode field uses M bits to address 45 operations, each register operand uses N bits and the remaining bits are used for the immediate operand field.

Determine the minimum number of bits required for the Opcode field.

ii(b). A RISC processor has a 32-bit architecture, with instructions that are 1-word long. The processor has 64 registers each of which is 32 bits long and supports 45 instructions (operations). Besides the operator field which holds the operation code (Opcode), each instruction has an immediate operand and two register operands. The Opcode field uses M bits to address 45 operations, each register operand uses N bits and the remaining bits are used for the immediate operand field.

Determine the minimum number of bits required for each register operand field.

ii(c). A RISC processor has a 32-bit architecture, with instructions that are 1-word long. The processor has 64 registers each of which is 32 bits long and supports 45 instructions (operations). Besides the operator field which holds the operation code (Opcode), each instruction has an immediate operand and two register operands. The Opcode field uses M bits to address 45 operations, each register operand uses N bits and the remaining bits are used for the immediate operand field.

Determine the maximum number of bits available for the immediate operand field.

ii(d). A RISC processor has a 32-bit architecture, with instructions that are 1-word long. The processor has 64 registers each of which is 32 bits long and supports 45 instructions (operations).

State two key characteristics of this processor's instruction set.

iii(a). Most conventional computers have a SISD architecture.

What does SISD stand for?

iii(b). Most conventional computers have a SISD architecture.

Draw a block diagram that illustrates the SISD machine architecture.

Exercise 3 (17 marks)

i(a). With reference to operating systems, distinguish between paging and segmentation.

i(b). With reference to operating systems, state one important relationship between the operating system and computer hardware.

i(c). With reference to operating systems, describe the terms Real Time processing and Batch processing using real life scenarios as examples to demonstrate their application.

ii(a). Consider the following process handled in an operating system with the associated units of arrival and service times:

PROCESS	ARRIVAL TIME	SERVICE TIME
P1	0	7
P2	2	4

PROCESS	ARRIVAL TIME	SERVICE TIME
P3	4	1
P4	5	4

Using the Shortest Remaining Time First scheduling policy draw a Gantt chart for these processes.

ii(b). Consider the following process handled in an operating system with the associated units of arrival and service times:

PROCESS	ARRIVAL TIME	SERVICE TIME
P1	0	7
P2	2	4
P3	4	1
P4	5	4

Using the Shortest Remaining Time First scheduling policy, calculate the average waiting time and turnaround time for the processes.

ii(c). Consider the following process handled in an operating system with the associated units of arrival and service times:

PROCESS	ARRIVAL TIME	SERVICE TIME
P1	0	7
P2	2	4
P3	4	1
P4	5	4

Using the Shortest Remaining Time First scheduling policy, calculate the throughput of the operating system.

Exercise 4 (17 marks)

i(a). What is a Computer Network?

i(b). State and explain THREE advantages and TWO disadvantages of using Computer Networks.

ii(a). A new company has been established. It bought some new premises which consist of a number of buildings on a single site. It decided that all of the computer workstations in the different buildings need to be networked. The company's technical services are considering ways in which the network might be set up.

One option considered is to use cabling for the network and to install it themselves.

Name the three types of cabling that they might consider.

ii(b). A new company has been established. It bought some new premises which consist of a number of buildings on a single site. It decided that all of the computer workstations in the different buildings need to be networked.

One option considered is to use cabling for the network and to install it themselves.

Explain two factors, other than cost, that they need to consider when choosing suitable cabling.

ii(c). A new company has been established. It bought some new premises which consist of a number of buildings on a single site. It decided that all of the computer workstations in the different buildings need to be networked.

Another option considered is to use wireless technology for at least part of the network.

Explain one type of wireless network that might be suitable for wireless networking.

ii(d). A new company has been established. It bought some new premises which consist of a number of buildings on a single site. It decided that all of the computer workstations in the different buildings need to be networked.

Another option considered is to use wireless technology for at least part of the network.

Identify one advantage, other than cost, of using wireless rather than cable networking.

ii(e). A new company has been established. It bought some new premises which consist of a number of buildings on a single site. It decided that all of the computer workstations in the different buildings need to be networked.

Another option considered is to use wireless technology for at least part of the network.

Identify one disadvantage (other than cost) of using wireless rather than cable networking.

Exercise 5 (17 marks)

i. Poorly planned, un-normalised databases can suffer from various anomalies when we manipulate the data. Describe two types of anomalies that can occur in a database, giving an example of each.

ii(a). Geography teachers at Bright Future High School organize several field trips throughout the year. They may organize more than one trip in a year and students may book and pay for more than one trip. A database has been created to record student payments. The entities and attributes are as follows:

STUDENT(Student_ID, Name, Gender, Class)
TRIP(Trip_ID, Trip_Leader, Destination, Date, Cost)
PAYMENT(Student_ID, Trip_ID, Payment_Date, Amount)

Define the terms *primary key*, *composite key* and *associative table*, giving an example of each from this database.

ii(b). Geography teachers at Bright Future High School organize several field trips throughout the year. They may organize more than one trip in a year and students may book and pay for more than one trip. A database has been created to record student payments. The entities and attributes are as follows:

STUDENT(Student_ID, Name, Gender, Class)
TRIP(Trip_ID, Trip_Leader, Destination, Date, Cost)
PAYMENT(Student_ID, Trip_ID, Payment_Date, Amount)

State and explain the cardinality ratio of the relationship that exists between STUDENT and TRIP.

ii(c). Geography teachers at Bright Future High School organize several field trips throughout the year. They may organize more than one trip in a year and students may book and pay for more than one trip. A database has been created to record student payments. The entities and attributes are as follows:

STUDENT(Student_ID, Name, Gender, Class)
TRIP(Trip_ID, Trip_Leader, Destination, Date, Cost)
PAYMENT(Student_ID, Trip_ID, Payment_Date, Amount)

Write an SQL query that displays for each trip, the number of payments made. The number of payments should be displayed in a column labeled Number_of_Payments.

Exercise 6 (17 marks)

i(a). Consider the arithmetic expression $E = A + B \times C / D$.

With respect to arithmetic expressions, define the terms infix, prefix and postfix.

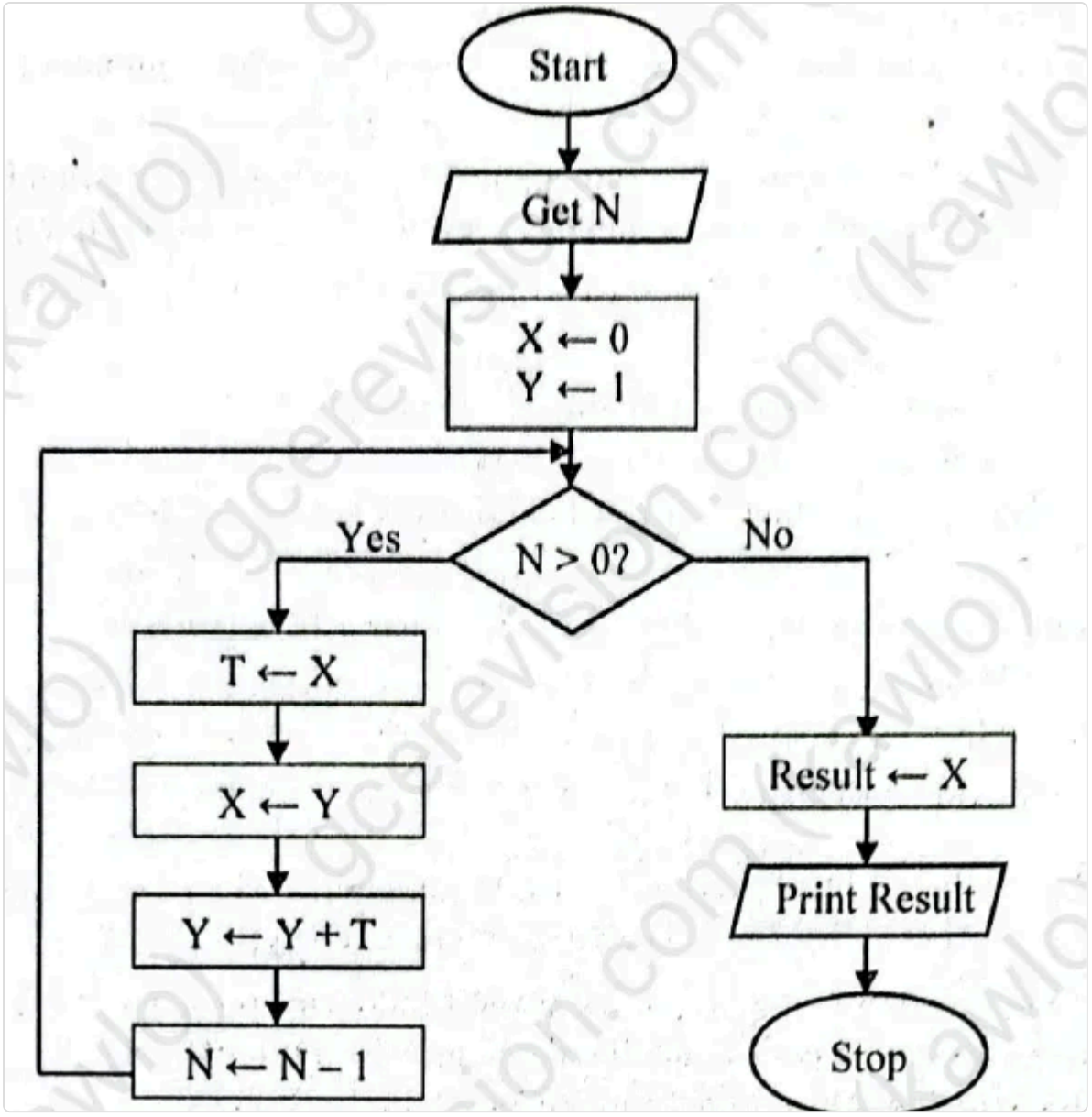
i(b). Consider the arithmetic expression $E = A + B \times C / D$.

Give the expression E stored as an expression tree, in the order read.

i(c). Consider the arithmetic expression $E = A + B \times C / D$, stored as an expression tree with $+$ at the root, A as its left child and the subtree for $(B \times C)/D$ as its right child.

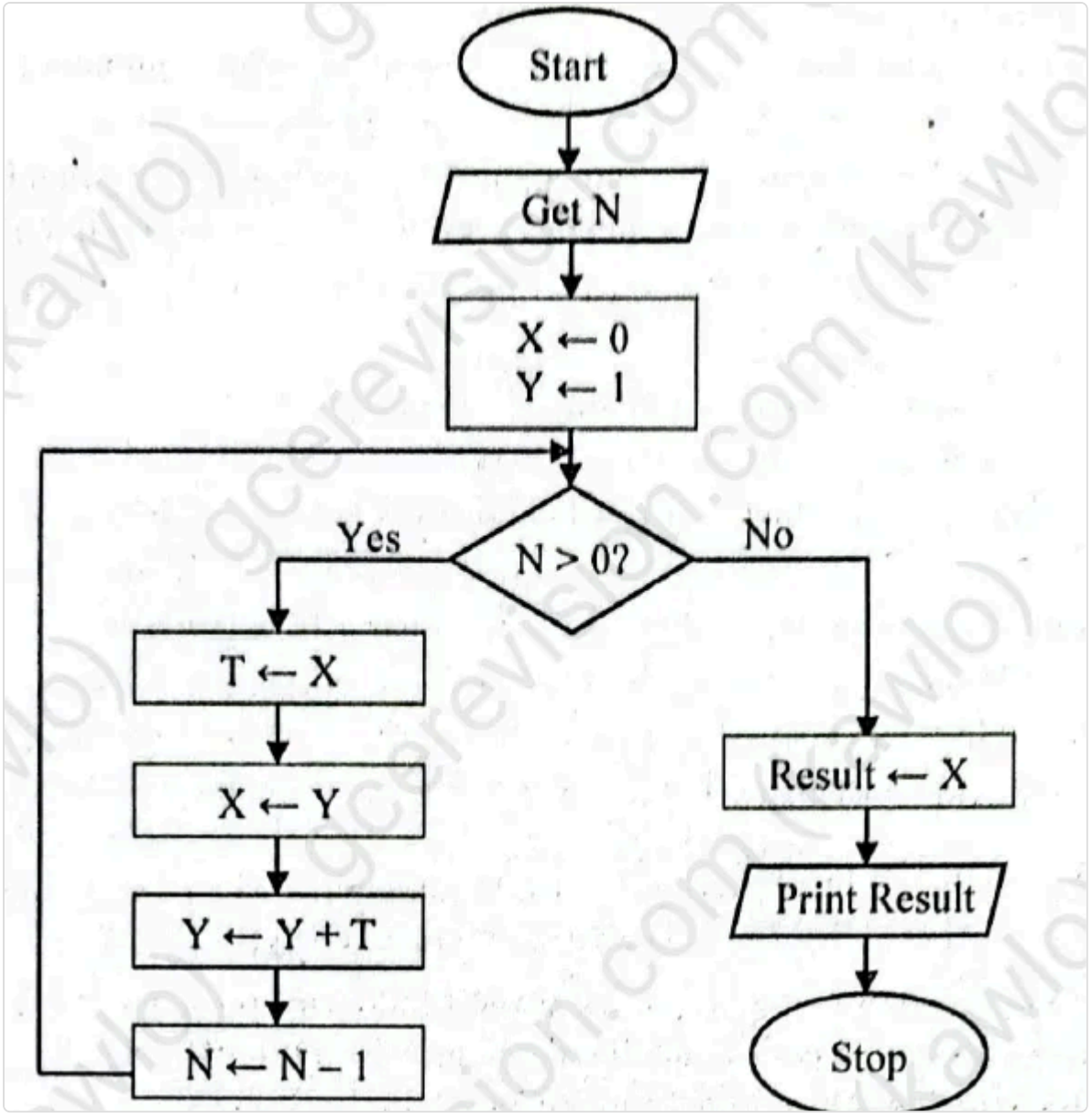
State in words how you would obtain the expression E in postfix form from the expression tree. Give the postfix expression obtained.

ii(a). Study the following flowchart algorithm and answer the questions that follow.



Give a trace table of the values for the variables, X, Y and N at the point where N>0 is tested, where N is initially 8. Hence, determine the output of this algorithm for N = 8.

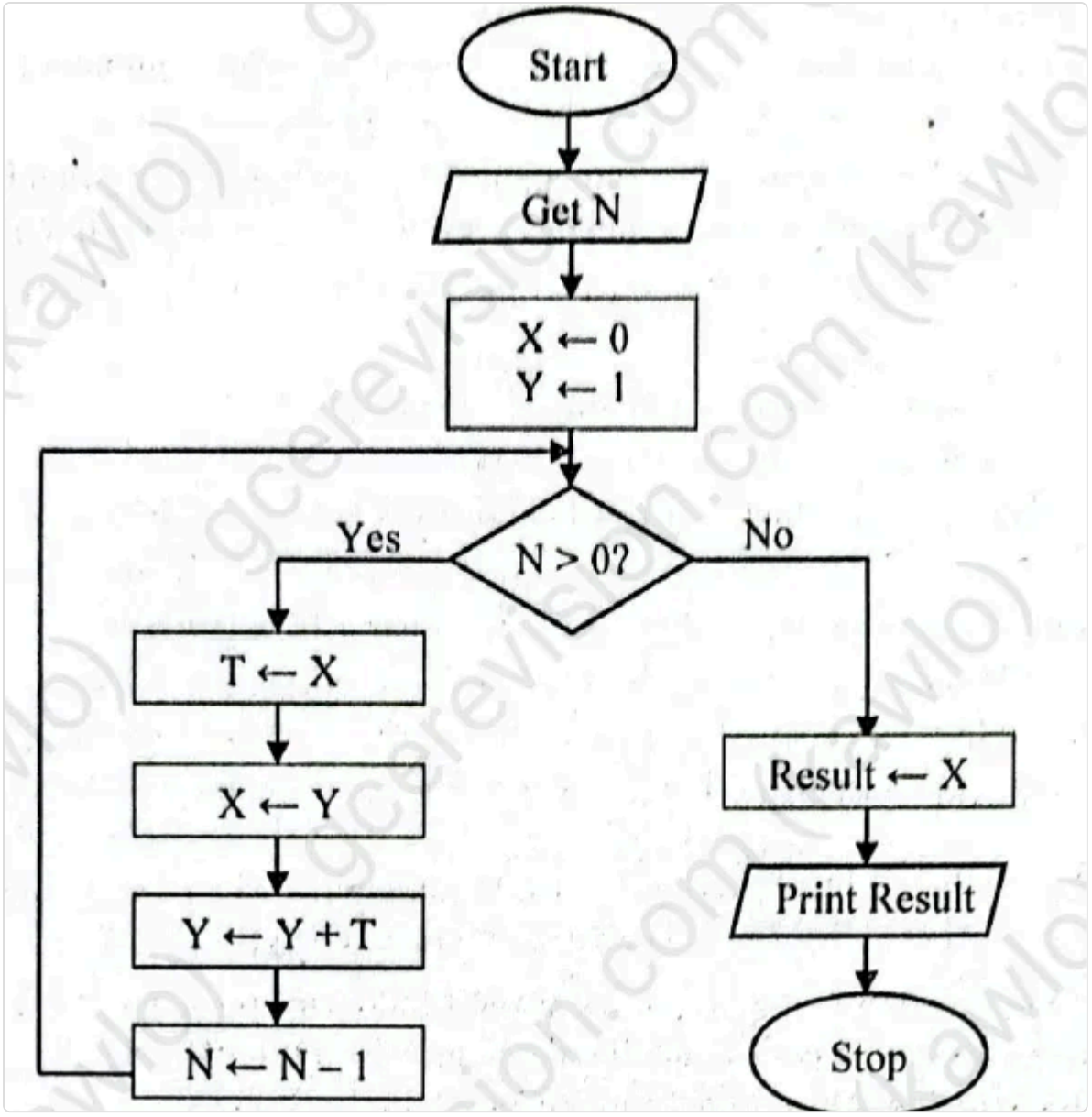
ii(b). The flowchart algorithm below is equivalent to this pseudocode (the flowchart is shown in the figure).




```
Start
Get N
X <- 0
Y <- 1
WHILE N > 0
    T <- X
    X <- Y
    Y <- Y + T
    N <- N - 1
ENDWHILE
Result <- X
Print Result
Stop
```

From your trace table in (a) above, deduce the computation that is performed by the above algorithm. Write a mathematical formula for this computation.

ii(c). The flowchart algorithm below is equivalent to this pseudocode (the flowchart is shown in the figure). It reads N and loops while N > 0.




```
Start
Get N
X <- 0
Y <- 1
WHILE N > 0
    T <- X
    X <- Y
    Y <- Y + T
    N <- N - 1
ENDWHILE
Result <- X
Print Result
Stop
```

State the time complexity of the above algorithm.

Exercise 7 (17 marks)

i(a). What is a program control construct?

i(b). With the help of flow diagrams, illustrate the difference between a pre-test loop and a post-test loop.

i(c). Give one example each of pre-test loop and post-test loop.

ii(a). Consider the functions f1 and f2 below that perform the same task.

```
1. FUNCTION f1(m,n) ≡
2.   IF(n=0)
3.     return 0
4.   ELSE
5.     return m + f1(m,n-1)
6. END
```

```
1. FUNCTION f2(m,n) ≡
2.   f = 0
3.   WHILE(n > 0)
4.     f = f + m
5.     n = n-1
6.   ENDWHILE
7.   return f
8. END
```

For m = 5 and n = 4, determine the output for any one of the above algorithms.

ii(b). Consider the functions f1 and f2 below that perform the same task.

```
1. FUNCTION f1(m,n) ≡
2.   IF(n=0)
3.     return 0
4.   ELSE
5.     return m + f1(m,n-1)
6. END
```

```
1. FUNCTION f2(m,n) ≡
2.   f = 0
3.   WHILE(n > 0)
4.     f = f + m
5.     n = n-1
6.   ENDWHILE
7.   return f
8. END
```

Which of the above algorithms is more efficient? Explain.

ii(c). Consider the functions f1 and f2 below that perform the same task.

```
1. FUNCTION f1(m,n) ≡
2.   IF(n=0)
3.     return 0
4.   ELSE
```

```
5.     return m + f1(m,n-1)
6. END
```

```
1. FUNCTION f2(m,n) ≡
2.   f = 0
3.   WHILE(n > 0)
4.     f = f + m
5.     n = n-1
6.   ENDWHILE
7.   return f
8. END
```

Explain the difference between an argument and a parameter.

Exercise 8 (17 marks)

i. Some of the steps in computerising an existing manual system include: systems analysis, systems design, programming, testing, changeover to the new system, operation and maintenance.

Briefly describe two tasks which will be performed during the design process.

ii. Some of the steps in computerising an existing manual system include: systems analysis, systems design, programming, testing, changeover to the new system, operation and maintenance.

Explain how possible it is for all the individual component modules to pass their tests yet the system fails to run.

iii. Some of the steps in computerising an existing manual system include: systems analysis, systems design, programming, testing, changeover to the new system, operation and maintenance.

The changeover to the new system from the manual system can be achieved in three ways:
(a) Immediate change;
(b) Running the manual and computerised systems in parallel;
(c) Gradually introducing the new system a subsystem at a time.

For each of the changeover techniques above, state an application which is most appropriate.

iv(a). An important part of any project is testing. Two testing techniques are white box testing and black box testing.

Explain the difference between them.

iv(b). Consider the procedure below that decides whether or not a person can vote, depending on their age. The minimum voting age is 18.

```
PROCEDURE CanVote(age)
    IF age >= 18 THEN
        OUTPUT *"you can vote"*
    ELSE
        OUTPUT *"you cannot vote"*
    ENDIF
END PROCEDURE
```

Design suitable test data that will test the procedure adequately. Justify your choice of data item(s).

Want to check your answers?

The full correction of this paper, with the method explained step by step, is available on EduKora: 15,000 FCFA per child until the exam, 12,000 FCFA for each additional child in the same family.

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